## 6.6 Induction type wattmeter

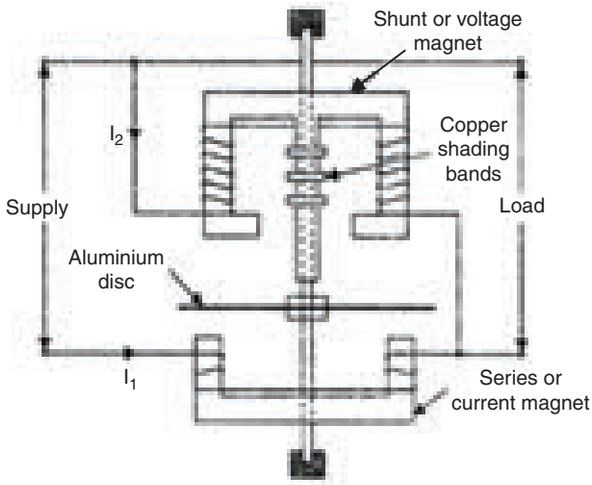



**Fig. 6.10 Induction type wattmeter**

Figure 6.10 shows an induction type wattmeter. It has two laminated electromagnets, namely shunt magnet and series magnet. The shunt magnet is excited by pressure coil and carries current which is proportional to the voltage of the circuit. Its pressure coil is connected in parallel with the circuit. The series magnet is excited by current coil which is connected in series with the load. A thin aluminium or copper disc is mounted in such a way that it cuts the fluxes of both the magnets.

Hence two eddy currents are produced in the disc. The deflection torque is produced due to the interaction of these eddy currents and the inducing fluxes. Two or three copper rings, called shadings rings are fitted on the central limb of the shunt magnet and can be so adjusted as to make the resultant flux in the shunt magnet lag behind the applied voltage by exactly 90°.

This instrument is provided with spiral springs to provide controlling torque. The spring being fitted to the spindle of the moving system and carries a pointer. The scale is quite uniform and extends over 300°. Induction watt meters can be used on AC circuit only.

### a) Advantages

i. Fairly long scale. (extending over 300°)
ii. Free from the effects of stray fields.
iii. Good damping.
iv. Practically free from frequency errors.

### b) Disadvantages

i. Sometimes subject to serious temperature errors.
ii. Power consumption is comparatively higher.
iii. Heavy moving system.

## 6.7 Multimeter

A multimeter is used to measure the following quantities.

i. AC voltage and current of different ranges
ii. DC voltage and current of different ranges
iii. Resistance of different ranges

As multi meter is used for measuring current (A), voltage (V) and resistance (Ohm), it is also called AVO meter. There are two types of multimeters, namely analog multimeter and digital multimeter.

An analog multimeter is basically a permanent magnet moving coil galvanometer. The scale is calibrated for each range and type of measurement. The range and the particular type of measurement are selected by a selector switch.



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